

CUPS 0day (Information Security Laboratory)

CUPS 0day (Information Security Laboratory) - Adrian Pastor, lab.gnucitizen.org.

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Content

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DESCRIPTION:

The CUPS daemon (/usr/sbin/cupsd) which listens by default on port 631/tcp, crashes when more than 100 RSS Subscriptions are added. *No authentication* is required to perform such action on CUPS <1.3.8. The caveat is that by default - at least on Ubuntu and openSuse - the daemon only accepts connections from localhost as specified by the default configuration settings (/etc/cups/cupsd.conf). However, the attack can be of **remote** nature by tricking the victim user to visit a **specialty-crafted page**. Such page would forge the 'add rss subscription' request 101 times which causes the CUPS daemon to crash.

The CUPS daemon *runs by default* on Ubuntu, openSuse and probably other GNU/Linux distributions. Additionally, this vulnerability can be replicated against CUPS daemons using *default settings*. Since no authentication is required to add new RSS subscriptions on CUPS <1.3.8, the CUPS administrator does **not *need* to be logged in during exploitation.

PoC TESTED ON:

Ubuntu 8.04.1 (fully patched as of 19th Oct 2008) Linux 2.6.24-21-generic #1 SMP Mon Aug 25 17:32:09 UTC 2008 i686 GNU/Linux

openSUSE 11.0 (i586) Linux 2.6.25.5-1.1-default #1 SMP 2008-06-07 01:55:22 +0200 i686 i686 i386 GNU/Linux

Common UNIX Printing System 1.3.7 Remote DoS PoC tested on Mozilla Firefox 3

Other Linux distributions are also suspected to be vulnerable but have NOT been tested.

24th Nov 2008 UPDATE:

There are three separate issues that would allow an "evil" webpage to cause your CUPS daemon to crash, even if you are *not *logged into the CUPS web interface:

- CUPS allows anonymous users to add/remove RSS Subscriptions. This issue only affects CUPS <1.3.8. I later learned that this issue had been reported in the past and tracked by Apple as [STR #2774](#). This issue is also being tracked as [CVE-2008-5184](#)
- HTTP requests submitted to the CUPS web interface (<http://localhost:631/>) can be forged due to lack of tokenization (CSRF)
- Exceeding the maximum # of RSS Subscriptions (100 by default) leads to a NULL pointer dereference crash. This issue is being tracked as [CVE-2008-5183](#)

22th Nov 2008 UPDATE:

If you're running CUPS >= 1.3.8, then this crash is mitigated by the fact that the victim user needs to be logged into the CUPS web interface for the crash to occur. This issue was addressed by Apple as STR #2774: <http://www.cups.org/str.php?L2774> <http://www.opensource.apple.com/darwinsource/Current/cups-136.11/cups/CHANGES.txt>

So for instance, if you are running Ubuntu 8.10 (intrepid), then the crash only happens if you're logged into CUPS. On the other hand, no authentication is required on Ubuntu 8.04.1 LTS (hardy) to crash cupsd.

Either way, the fact that the daemon crashes when the maximum number of RSS subscriptions is exceeded is a* separate issue* to the lack-of-authentication problem when managing RSS subscriptions (<1.3.8). Even though on Mac OS X Leopard (CUPS >=1.3.8) authentication *is* required to manage RSS subscriptions, Apple has acknowledged the crash issue and is currently working on a fix.

Finally, this bug cannot lead to code execution, since it's a NULL pointer dereference as confirmed by redb0ne and Josh Bressers from Red Hat Security Response Team: <http://openwall.com/lists/oss-security/2008/11/19/4>**

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- [cups_dos_poc.html.txt](#) 1k - on Nov 19, 2008 2:48 PM by Adrian Pastor (version 2 / [earlier versions](#))